Royal Crest Academy
SCIENCE · INNOVATION · SKILL

FREE LEARNER GUIDE

Mathematics *study guide*

Mathematics trains you to reason clearly, spot patterns and solve problems, and it is the tool that every science and technical subject depends on.

Big ideas

Number and algebra

Algebra lets one rule describe a whole pattern. Solving an equation means undoing its operations in the right order.

Shape and space

Geometry, trigonometry and Pythagoras' theorem ($a^2 + b^2 = c^2$) let us measure lengths and angles we cannot reach.

Data and chance

Statistics summarises information, and probability measures how likely an event is, from 0 (impossible) to 1 (certain).

Topic checklist

tick each topic when you can explain it

O LEVEL

- [ ] Number, fractions and percentages
- [ ] Ratio, rates and proportion
- [ ] Indices and standard form
- [ ] Algebraic expressions and formulae
- [ ] Equations and inequalities
- [ ] Graphs of functions
- [ ] Sets and Venn diagrams
- [ ] Geometry and angle properties
- [ ] Mensuration: area and volume
- [ ] Trigonometry and Pythagoras
- [ ] Vectors and matrices
- [ ] Statistics and probability

A LEVEL

- [ ] Quadratics and functions
- [ ] Coordinate geometry
- [ ] Sequences and series
- [ ] Differentiation
- [ ] Integration
- [ ] Trigonometry
- [ ] Exponentials and logarithms
- [ ] Vectors
- [ ] Mechanics
- [ ] Probability and statistics

Key terms

Integer	A whole number, which can be positive, negative or zero.
Prime number	A whole number greater than 1 with exactly two factors: 1 and itself.
Coefficient	The number in front of a letter in algebra, such as the 3 in $3x$.
Gradient	The steepness of a line: the change in y divided by the change in x.
Hypotenuse	The longest side of a right-angled triangle, opposite the right angle.
Mean	The sum of the values divided by the number of values.
Probability	A number from 0 to 1 that measures how likely an event is.
Function	A rule that maps each input to exactly one output.

Skills you will build

- Mental arithmetic and estimation
- Using a scientific calculator accurately
- Constructions with ruler, compasses and protractor
- Drawing and reading graphs
- Setting out clear, logical working
- Checking answers by substitution

Study smarter

1. Practise a little every day rather than a lot once a week.
2. Learn the formulae and when to use each one.
3. Always show your working: method marks count.
4. Check an answer by putting it back into the question.

Check yourself

1. Solve $3x + 5 = 20$.

2. A right-angled triangle has shorter sides of 6 cm and 8 cm. How long is the hypotenuse?

3. Find the mean of 4, 7, 9 and 12.

Answers 1. $x = 5$ (subtract 5 to get $3x = 15$, then divide by 3). · 2. 10 cm ($6^2 + 8^2 = 36 + 64 = 100$, and $\sqrt{100} = 10$). · 3. 8 ($4 + 7 + 9 + 12 = 32$, and $32 \div 4 = 8$).

Where it leads: Engineering, Actuarial science, Accounting and finance, Data science, Architecture, Teaching